Given these sample points:

[0.31940389 0.76295937]
[0.57432921 0.8798981]
[0.73102363 0.73299988]
[0.84035342 0.26473161]
[0.65011406 0.68152635]
[0.41043714 0.1475543]
[0.31269116 0.07872278]
[0.68341817 0.86105746]
[0.08250725 0.40348751]
[0.88388983 0.58225397]
[0.908656 0.808513]
[0.43 0.5]
[0.611853 0.139494]
[0.889128 0.578807]
[0.4 0.7]
[0.7 0.45]
[0.807957 0.767765]
[0.735 0.735]

And these results:

1.3226770395454077e-79
1.0330782375230975e-46
7.710875114502849e-16
3.341771007676023e-124
-0.0036060626443634764
-2.1592490357331095e-54
-2.0890932702320842e-91
2.5350011535584046e-40
3.6067711901420254e-81
6.229856468168659e-48
2.358757555918235e-78
-4.336733960179099e-05
2.4247688654824617e-82
5.479582068424992e-50
2.653959939729813e-40
-2.5135190519902422e-26
1.3771514819813648e-36
6.824045110810138e-17

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperature = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially doing **black-box optimization** with sampled points, and your goal is to pick the next point likely to give a higher value.

---

1. Identify the current best region

From your results:

- The **largest value** is:

```
7.710875114502849e-16
```

  at:

```csharp
[0.73102363, 0.73299988]
```

- The next closest good value:

```
6.824045110810138e-17
```

  at:

```csharp
[0.735, 0.735]
```

👉 These are clearly **clustered around (~0.73, ~0.73)**

---

2. Interpret the landscape

- Most other values are **extremely small (≈0) or negative**
- There is a **localized peak near (0.73, 0.73)**
- The function likely has a **sharp maximum** in that region

---

3. Strategy (given constraints)

You said:

- Don't sample the same point twice ✅
- Low exploration (Temperature = 0.1)
- Top-k = 3, Top-p = 3 → **very exploitative**

👉 So the model should **refine locally near the best point**, not explore far away.

---

4. Best next sample

You want to **probe very close to the current maximum**, but not duplicate.

Good candidates (slight perturbations):

- **Primary recommendation:**

```csharp
[0.74, 0.74]
```

- Other strong alternatives:

```csharp
[0.72, 0.74]
[0.74, 0.72]
```

---

5. Why this works

- You are effectively doing **local exploitation**
- The peak appears smooth and centered near (0.73, 0.73)
- Small moves help:
  - Confirm the peak location
  - Potentially climb slightly higher

Final Answer

👉 **Sample next at:**

```csharp
[0.74, 0.74]
```

This is the most likely point to yield a higher value under your constraints.

If you want, I can also suggest a **Bayesian optimization strategy** or fit a surrogate model to pick the mathematically optimal next point.